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Innovative Application of Acoustic Emission Monitoring of Multiphase Flow in Intelligent Completions

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Abstract

Pioneering testing procedures and a well-defined laboratory setup enabled studying the application of acoustic emission (AE) techniques in characterizing multiphase flows within intelligent completions. The study provided a comprehensive understanding of the testing methodologies and experimental conditions required for successful implementation.

Amongst other applications of acoustic sensing in both upstream and downstream oil and gas, the focus in this laboratory study was to assess the feasibility of passive acoustic techniques in real-time detection of flow-induced noise (FIN) within downhole completions, specifically targeting Venturi throat and inflow control devices (ICDs). The scope encompassed evaluation of single-phase flows (water and oil) and two-phase flows (oil-gas and oil-water) at different flow rates with azimuthal recording around the flow restriction.

The laboratory study involved a series of flow loop tests simulating multiphase flows in downhole completion equipment. Miniature AE sensors (hydrophones) were strategically placed near flow restrictions, capturing FIN. The encountered challenges with implementation for such a complex testing setup required innovative and engineered solutions. The acoustic signals' energy was then correlated with changes in fluid flow properties. The testing procedures utilized a customized setup with acoustic sensors to obtain optimal sensor placement for effective monitoring.

Test results indicated successful study of correlations between the total flow rate, water cut, and free gas cut with the energy and spectral characteristics of the acquired FIN acoustic signal. The laboratory setup, equipped with miniature hydrophones and a signal conditioner, facilitated the acquisition of accurate acoustic signals. Observations point to the potential of using FIN for early detection of water and gas breakthrough, enhancing real-time production monitoring capabilities. The testing procedures demonstrated the feasibility of employing an acoustic-based permanent downhole flow monitoring system.

This work contributes novel insights into AE techniques, particularly in the context of intelligent completions. The innovative laboratory setup offers a pioneering approach to assessing flow-induced noise. The insights gained from the laboratory testing contribute to the existing body of literature and offer practical

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guidance to practicing engineers for efficient and accurate acoustic-based flow monitoring in the petroleum industry.

Introduction

Advancements in well construction have enabled placing long horizontal multilateral wells as a viable technology to maximize reservoir contact. However, these wells come with their own challenges in achieving optimal and uniform reservoir drainage due to reservoir heterogeneity and heel-to-toe effect which may cause unwanted water coning or gas breakthrough. To maximize hydrocarbon recovery and lower field development costs, the industry has been moving towards intelligent completions in which horizontal wellbores are segmented into isolated producing compartments, each equipped with permanent downhole system to monitor and control the fluid flow in real-time.

By design, many downhole components of intelligent completions include fluid flow restrictions, such as Venturi throats in permanent downhole flowmeters and nozzles of inflow control devices (Ellis, T. et al. 2010). The objective of the present work is to assess the feasibility of utilizing passive acoustic techniques in real-time characterization of multiphase flows in downhole intelligent completions by acquiring and analyzing flow induced acoustic noise (FIN) generated at these flow restrictions.

In this paper, we present a laboratory study in which single-phase (oil or water) as well as two-phases (oil-gas an oil-water) flows are reproduced in the test section of the flow loop to simulate downhole flow conditions through real-size Venturi throat and ICD nozzles. FIN is acquired using passive acoustic sensors (hydrophones) installed near the flow restrictions where it is generated. Energies and spectra of the obtained FIN signals were then analyzed for correlations with changes in fluid flow properties, such as total flow rate and phase volume fractions. The results show the potential of using FIN signal properties for early detection of water and gas breakthrough, thus enhancing real-time production monitoring capabilities. The testing procedures demonstrated the feasibility of employing a cost-effective acoustic-based permanent downhole flow monitoring system. This system, when coupled with an interval control valve (ICV) that allows automatic shut-off of a section of the well, will prevent unwanted fluids (water or gas) from entering the well before they are detected at the surface. Furthermore, the acquired acoustic data from this study will help to train and test deep neural networks for flow regime identification in pipe flow, enabling the creation of a predictive model for real-time flow monitoring (Iqbal et al. 2025).

The paper includes detailed experimental setup and procedure, followed by presentation of experimental results and conclusions.

Experimental Setup

Test Equipment

Multiphase flows of oil, water and gas were simulated using a large 3-phase flow loop system. This system allows controllable circulation of multiphase fluid mixtures through the test section at the defined rates and phase volume fractions. Capacity of pumping and acquisition systems of the flow loop along with flexibility in customization of test section – enable full-scale testing of downhole completion equipment (Saeed et al. 2015).

Figure 1 shows the schematic of the 3-phase flow loop, as it was used in this work. Test section was built from transparent 4-inches ID PVC tubes to allow observations of the flows during the experiment (G. Falcone al. 2008). The specific flow restrictions, aimed to simulate real-size Venturi throat of downhole flowmeter or nozzle-type ICD – were designed and implemented as interchangeable plastic or acrylic glass modules, deployed inside the test section. Four (4) miniature acoustic sensors (hydrophones), placed at 90 degrees angle apart, were submerged inside the test section and coupled to the specific flow restriction module. The chosen number of hydrophones is for redundancy, as well as for evaluating anisotropic

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responses. Special care had to be taken to protect the coaxial signal cables of hydrophones as they got exposed to the test fluids by encapsulating them inside the oil-resistant flexible tubing all the way between the sensor and feedthrough on test section flange.

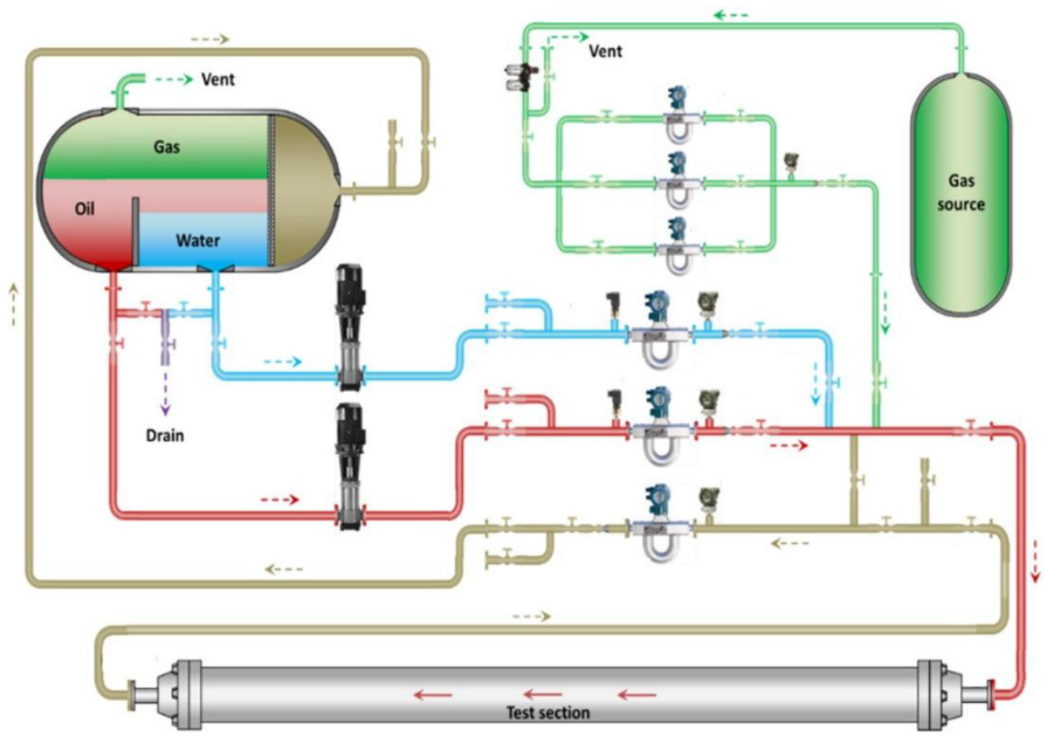

Figure 1—Schematic of the 3-phase flow loop system. Controlled flow of oil, water and gas enter as a multiphase flow mixture into the test section and return back to the separator.

General schematic of the test section is shown in the following Figure 2. Test section, made of transparent PVC tubing – has an overall length of 14 ft (4.3 m) and inner diameter (ID) of 4 in. There is an inner tubing referred to as production tubing, installed to simulate downhole completion, as shown with blue color in Figure 2. Multiphase fluid mixture enters the test section from inlet and then flows into the inner production tubing through real-size nozzle type ICDs. The implemented dimensions of the test section allow for the multiphase flow to attain the horizontal pipe flow regime before it reaches the inner production tubing, located away from perturbation at the test section inlet (Schaeffer et al. 2024).

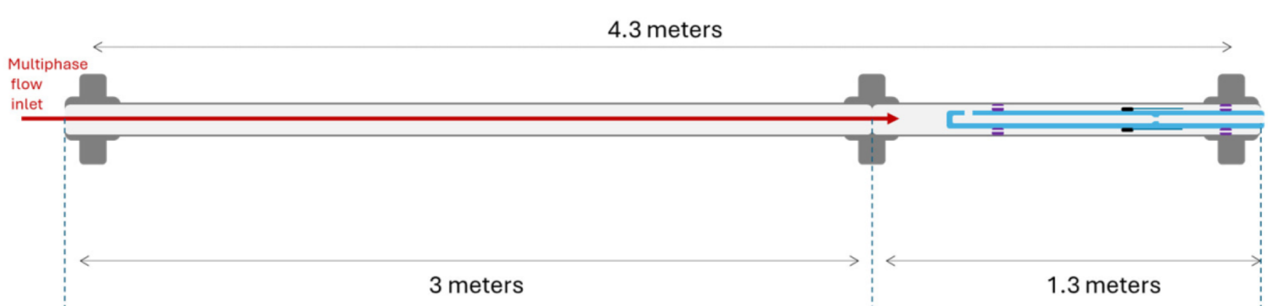

Figure 2—Test section schematic. As indicated with red arrow, multiphase flow mixture enters the test section from the inlet and reaches inner production tubing, highlighted with blue color. Production tubing directs the flow through the tested restrictions and further to the outlet.

Figure 3 shows an annotated photograph of the complete production tubing module configured to simulate FIN generated from the standard ICD nozzles. Production tubing has an adjustable length of about

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23 in. (600 mm), an outer diameter (OD) of 2-3/8 in. (61 mm) and inner diameter (ID) of 1.0 in. (25.4 mm). Production tubing is fixed inside the outer PVC housing tubing with plastic centralizers, located in the stagnant flow region downstream the ICDs.

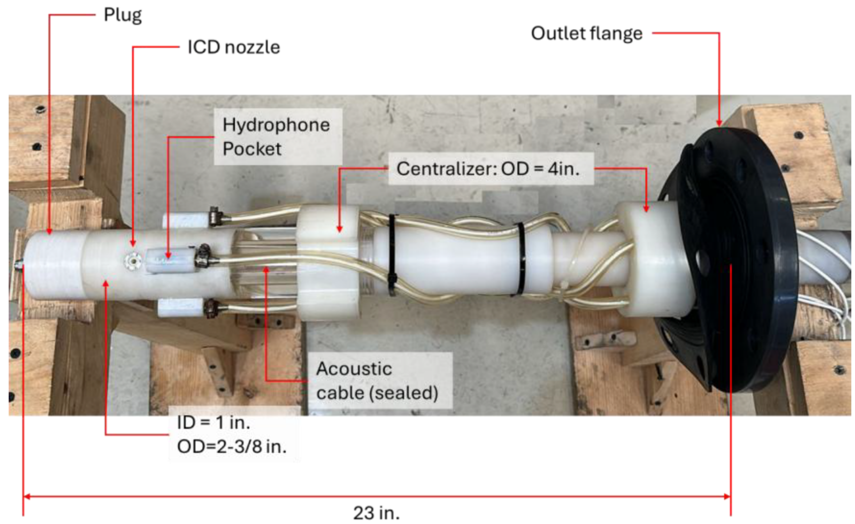

Figure 3—Photograph of the assembled production tubing prepared for installation inside the test section. Plastic ICD module is installed toward the tip of production tubing. Four (4) acoustic sensors (hydrophones) are installed inside the plastic seats (pockets) attached to the exterior of ICD module.

In this work, we studied FIN signal generated in configuration with standard ICD nozzles that are utilized in completion systems in the field, with orifice diameter of 4.0 mm. There are total of four (4) ICD nozzles staggered azimuthally by 90° within a single transverse plane relatively to the production tubing.

Figure 4 illustrates how ICD nozzles were installed in production tubing using the custom-built plastic module. Sensing tips of acoustic sensors (hydrophones), shown in Figure 4 with black color – are fixed tightly inside the plastic seats (pockets) adhered to the exterior of ICD module. In this way, each hydrophone is in the nearest proximity downstream of the corresponding ICD orifice.

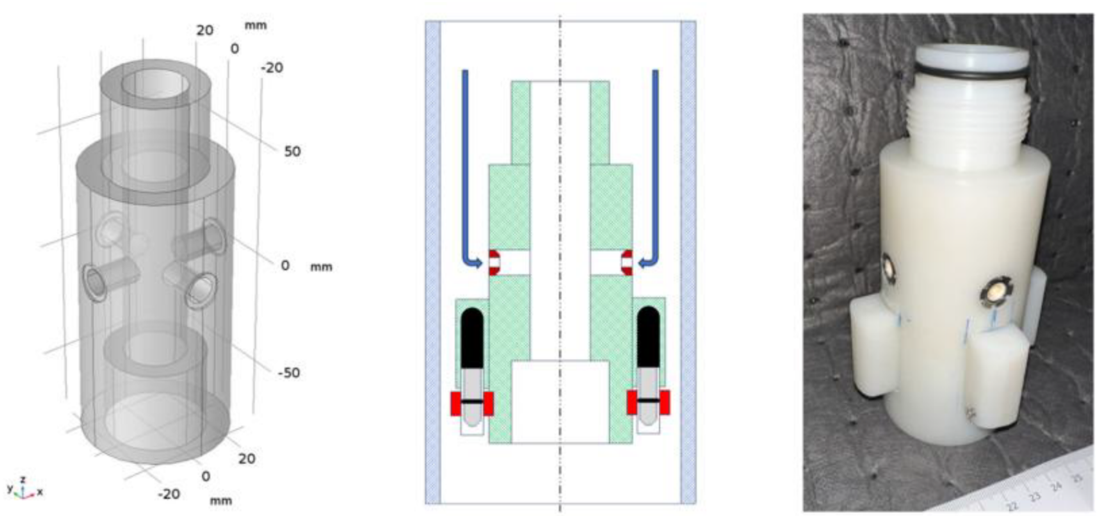

Figure 4—Left: design of the ICD plastic module. Center: schematic of the ICD module located inside the test section housing with blue arrows sowing direction of the annular flow into ICDs. Right: actual module with 4 ICD nozzles installed. Acoustic sensors are installed inside seats (pockets) glued to the module outer wall in the nearest proximity and downstream from individual ICD nozzle.

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The implemented 4×ICD configuration allowed flowrates up to 600 BPD, while remaining within test section maximum working pressure (MWP) of 40 psi. Table 1 summarizes main characteristics of the test section.

Table 1—Summary of test setup characteristics.

Test section outer tubing housing	Material	Clear PVC pipe, schedule 40
	Overall length	4.3 m
	ID	4.0 in.
	MWP	40 psi
Production tubing	Material	Acrylic glass; plastic
	OD	2.375 in.
	ID	1.0 in.
ICD module	Material	Plastic
	Inlet diameter	1.0 in.
	ICD orifice diameter	4.0 mm
	Quantity of ICDs installed	4, transversely to prod. tubing
Acoustic sensor	Make	Brüel & Kjær
	Model	8103, miniature hydrophone
	Frequency range	0.1 Hz to 180 kHz
	Quantity	4

Test Fluids

Desulfurized dearomatized hydrocarbon solvent D80 – widely available from various suppliers and hereafter briefly referred to as "D80 oil" – was used in this work as a substitute for light crude oil in flow loop testing. D80 and its analogues have been widely used as model oil in laboratory simulations of multiphase flows of oil, gas and water in downhole completion and surface production components (e.g., Saeed et al. 2015). Our benchtop checks confirmed immiscibility and fast separation of D80 and water mixture. Viscosity and density were found at 20°C near 2.9 cP and 0.82 g/ml, respectively.

Standard shop tap water was used as the aqueous phase in the flow loop experiments. The water in the separator was completely changed weekly to prevent the appearance of bacteria.

Air from the standard compressed air shop line was added into the multiphase flow, as gaseous phase. Flow loop metering skid is equipped with Coriolis mass flowmeters for gas flow which were utilized to setup the required flowrate of air. Air pressure filter regulator was used to adjust supply of air at the target rate and monitor for absence of condensate in the line.

Table 2 below summarizes fluids utilized in flow loop experiments.

Table 2—Fluids summary

Oil phase	Hydrocarbon solvent "D80"
Water phase	Standard shop tap water
Gas phase	Compressed air

Test Matrix

A test matrix was used to acquire the FIN signal generated at restrictions for various rates and phase volume fractions. Namely, the experiments were grouped by a certain fixed values of total flow rate of the generated

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2-phase flow through ICD restrictions, having oil as a continuous phase and either water or gas, as an added dispersed phase. For each value of total flow rate, volume fraction of the added phase was varied at predefined percentages: 0%, 10%, 30%, 50%, 70% and 100%. Selected values of the flowrates were demonstrated to cover a cavitation (600 BPD, hearable) and non-cavitating (300 and 450 BPD) conditions in the ICD.

In oil and water experiments, the required phase volume fractions of water, or water cut (WC) – was set up by controlling individual water and oil pumps, referring to the volume flowrate values (in BPD) reported by the flow loop acquisition system. In experiments with compressible gaseous added phase (air), air flow rate is set in terms of mass flow rate measured by Coriolis flowmeter and additional conversion from volumetric flowrate, or gas volume fraction (GVF) – is required. Given the total flow rate Q_{tot} and target GVF, mass flowrate, $\dot{m}_{air}$, of the required air supply into the test section is calculated as:

$$\dot{m}_{air} = \frac{0.158987}{24.0} \times GVF \times Q_{tot} \times \rho_{air}. \quad \text{(Eq. 1)}$$

Here the following symbols and unit conversion factors are utilized.

Table 3—Gas volume fraction calculation: notation.

Symbol/Constant	Unit	Note
$\dot{m}_{air}$	kg/hr	Mass flow rate of air supply into the test section
GVF	-	Gas Volume Fraction (GVF)
Q_{tot}	BPD	Total volumetric flowrate of oil and gas mixture 1 US Barrel = 0.158987 m3 1 Day = 24 hr
ρ_{air}	kg/m3	Mass density of air inside the test section.

In (Eq. 1) air density was corrected to the actual pressure and temperature conditions inside the test section. For that purpose, we utilized the average (stabilized) readings of inlet pressure, P_{in}, and outlet mixture temperature, T_{out}, during the flow. For the range of pressures and temperatures encountered in our tests, air density ρ_{air} was reliably estimated using the equation of state for the ideal gas.

Specific test matrices for flow loop experiments with ICDs – are given in Table 4 and Table 5 for the added phases of gas and water, respectively. For each total flow rate value, volume fraction of added phase – gas volume fraction (GVF) or water cut (WC) – was increased step-wise from 0% to 100%, simulating the scenario of gas or water breakthrough. Limiting case of GVF=100% correspond to only gas (air) flowing through the test section at the rate equal to the total rate of the series.

Table 4—Oil-Gas test matrix

Total Flowrate, BPD	300	450	600
Continuous Phase	Oil		
Dispersed Phase:Free Gas (% GVF)	0	0	0
	10	10	10
	30	30	30
	50	50	50
	70	70	70
	100	100	100

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Table 5—Oil-Water test matrix

Total Flowrate, BPD	300	450	600
Continuous Phase	Oil		
Dispersed Phase:Water (% WC)	0	0	0
	10	10	10
	30	30	30
	50	50	50
	70	70	70
	100	100	100

FIN Signal Analysis

Acoustic flow induced noise (FIN) is a known phenomenon in hydrodynamics. FIN is generated by small-scale pressure perturbations within the flow, such as turbulent and cavitation flows and can be encountered in

- Fluid flow through restrictions and near the side branches in piping systems;
- Flow leakage in pipeline.

In this paper, we acquire acoustic FIN signal generated in the lab simulation of multiphase flow through typical flow restrictions in downhole completion and study the relationship of FIN characteristics with fluid flow parameters, such as rate and phase volume fractions.

Each hydrophone in experimental setup provides a recording of acoustic signal S_n, sampled at discrete equidistant time instances t_n:

$$S_n = S(t_n), \qquad t_{n+t} = t_n + \Delta t. \tag{Eq. 2}$$

In all experiments reported here, FIN was acquired at 2000 Hz sampling rate for 120 seconds.

Time representation of the signal (Eq. 2) alone provides a limited insight in its characteristics. Hence, we used Fast Fourier Transform (FFT), a well-known and powerful signal processing approach, which is based on ability to represent any signal as a linear superposition of elemental harmonics of ascending frequencies:

$$S(t_n) = \frac{1}{N}\sum_{k=0}^{N-1} \hat{S}_k e^{i2\pi f_k t_n}, \qquad n = 0, 1, \ldots, N-1. \tag{Eq. 3}$$

Energy E_S of a discrete signal $S_n = S(t_n)$, $n = 0, 1, \ldots, N-1$; $N\Delta t = T$, acquired over the time interval $0 \le t \le T$ can be expressed using the Parseval's equality via signal's time and frequency representations:

$$E_S = \frac{\Delta t}{N}\sum_{k=0}^{N-1} \left|\hat{S}_k\right|^2 = \sum_{k=0}^{N-1} \Delta f \left|\Delta t \hat{S}_k\right|^2, \qquad \Delta f = \frac{1}{T}. \tag{Eq. 4}$$

This suggests defining the Energy Spectral Density (ESD) of discrete signal, as:

$$ESD\left(f_k\right) = \left|\Delta t \hat{S}_k\right|^2, \quad k = 0, \ldots, N-1. \tag{Eq. 5}$$

Having ESD plotted as function of ascending frequencies $f_k = k\Delta f$, $k = 0, 1, \ldots, N-1$, one obtains two-sided energy spectral density distribution, which is symmetrical. Due to this symmetry, in practice, one-sided energy spectral density is considered which uses frequencies f_k up to Nyquist limit, which is essentially half of the signal sampling frequency $f_s = \frac{1}{\Delta t}$.

In multiphase flow cases, such as water slugs or gas bubbles passing through ICD, acoustic FIN signals are nonstationary. To account for such cases, energy spectral density of the signal is usually calculated over short moving time windows, rather than over an entire acquisition duration. In this way, energy spectral

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density will be represented below as 2D colormaps, known as spectrograms, showing changes in spectral density over acquisition time.

To eliminate influence of duration of FIN recording, instead of overall energy of the signal, we look at its averaged value over the acquisition time, which we refer to as the signal power, P_S:

$$P_S = \frac{1}{T} E_S = \frac{1}{T}\int_0^T S^2(t)dt. \quad \text{(Eq. 6)}$$

In this work, we explored potential correlation between energy spectral density of the FIN signal generated at ICD module and multiphase flow characteristics, such as total flow rate and phase volume fraction, as well as dispersed phase type (water or gas).

Experimental Results

Oil-Gas Test Series

In this section, we present the sensitivities of acquired acoustic FIN spectral and energy properties observed in 2-phase oil & gas flow tests in ICD configuration.

Figure 5 demonstrates the dependence of average power of acoustic FIN signal P_S (Eq. 6) on the total flowrate ranging between 300 and 600 BPD and volume fraction of free gas (GVF) increasing from 0 to 100%. Here the dependence is presented for hydrophone #1 as the other three hydrophones demonstrated similar dependencies. Generally, FIN energy increases with flow rate for the complete range of GVF, except the 30% case which shown minimal and non-monotonous variation (Figure 5, left). It should be noted that single-phase flow at 600 BPD resulted in "hearable" cavitation noise at ICD, while 300 and 450 BPD did not produce hearable cavitation sound. Further, increasing GVF alone – does not change energy in monotonous way. Energy generally decreases once gas is added at 10% and then increases till it reaches its maximum at about 50% GVF and then decreases, as GVF reaches 70% and beyond (Figure 5, right).

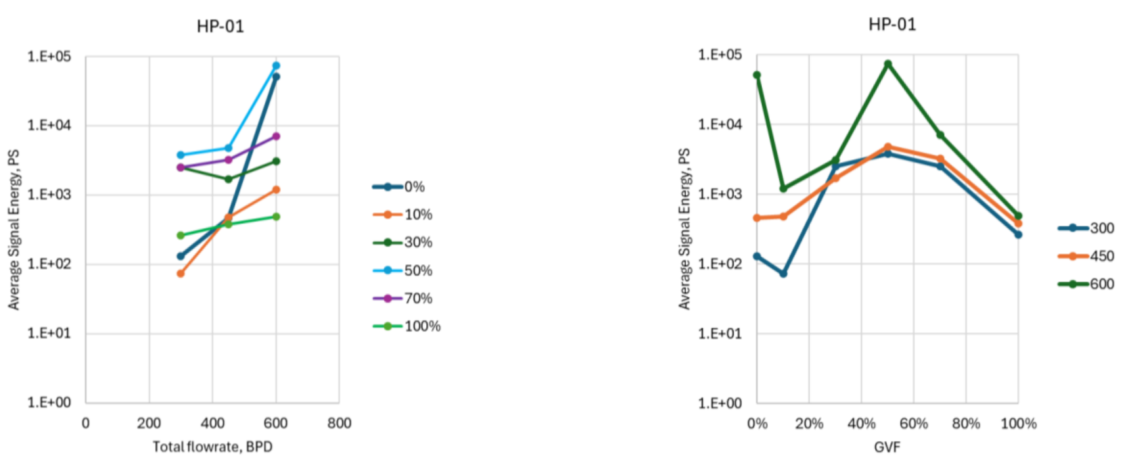

Figure 5—Oil-gas test series: correlation of acoustic FIN signal energy on total flowrate and free gas volume fractions (GVF).

Figure 6 presents spectrograms of acoustic FIN acquired for various flowrates and GVF by hydrophone #1, as the other three hydrophones revealed similar distributions and omitted here. Each spectrogram was computed using 330 milliseconds (ms) moving window, implying spectral resolution of 3 Hz. As soon as gas has been added in the flow, vertical bands appear in spectrograms. These bands (stripes) correspond to short time intervals when individual gas bubble is flowing through ICD nozzle.

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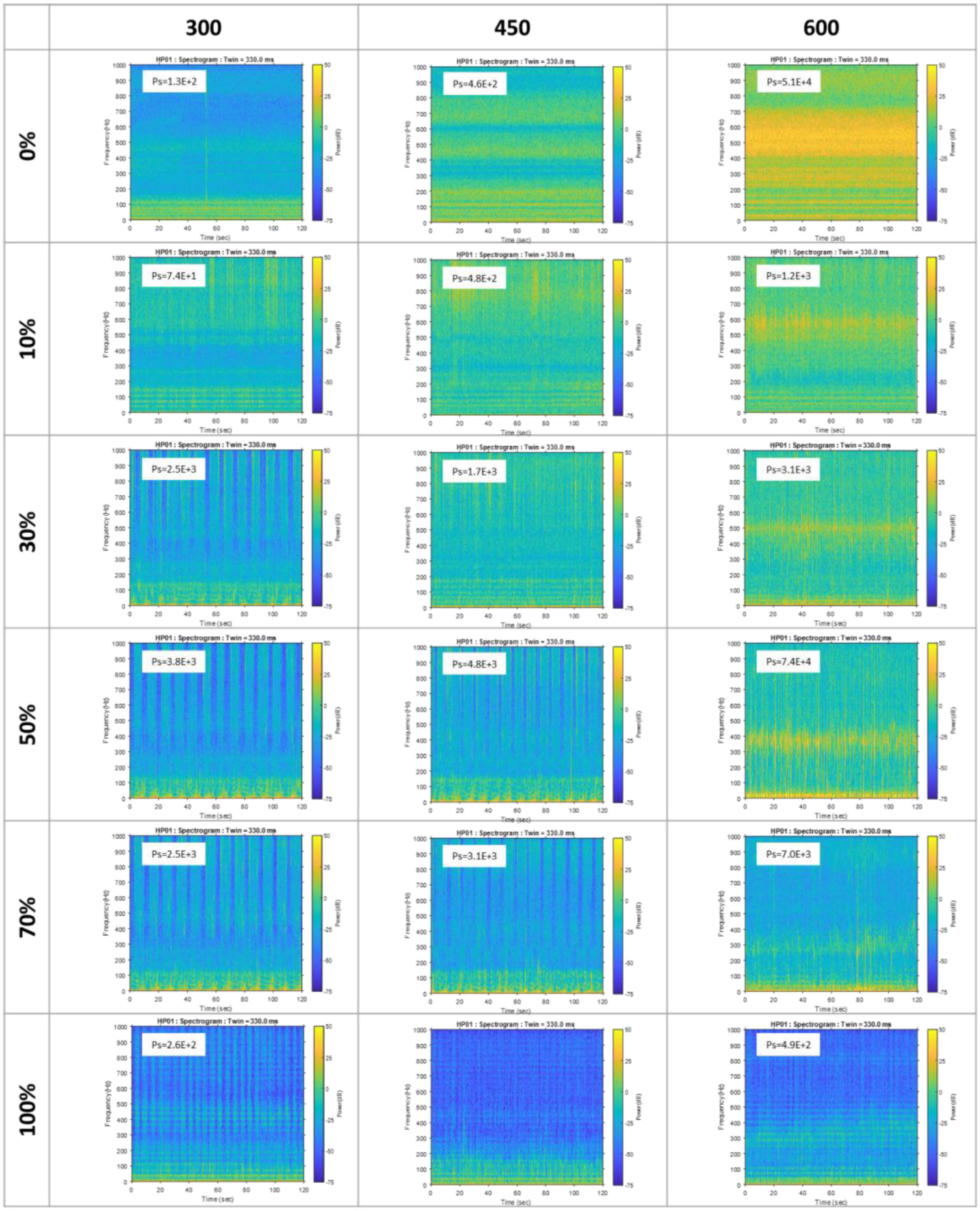

Figure 6—Oil-gas test series: spectrograms of acoustic FIN acquired at various total flowrates and GVF.

One can also notice horizontal bands present in the spectrograms for all GVF values. These horizontal bands correspond to certain frequencies of the strongest harmonics, which are present in the FIN signal continuously throughout the entire recording, except for short intervals when the signal was disturbed by a "gas bubble" passing through the ICD, represented in the spectrogram by the intersection with a vertical band. Positions of these frequency peaks vary with total flow rate and GVF. Figure 7 illustrates this by showing time-averaged spectrograms superimposed on each other.

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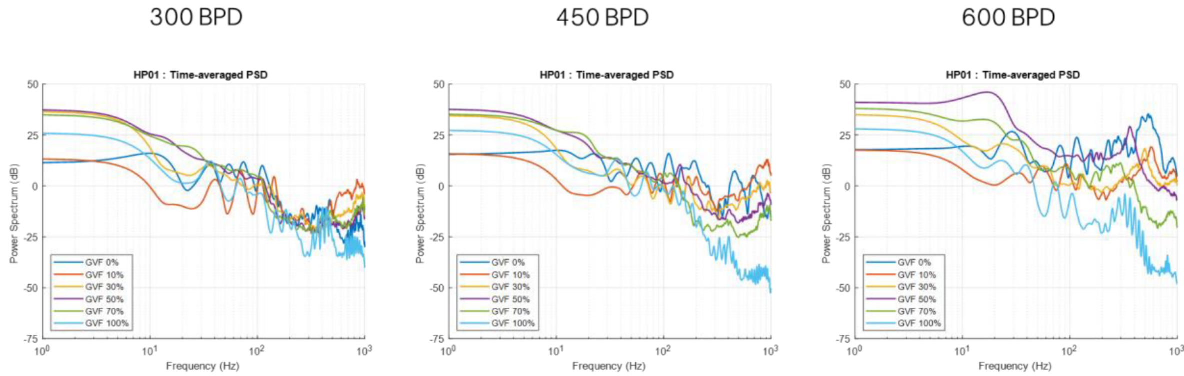

Figure 7—Oil-gas test series: comparison of time-averaged spectrograms for various rates and GVF.

Oil-Water Test Series

This section focuses on investigating how acoustic FIN spectral and energy properties acquired in 2-phase oil & water flow loop tests respond to characteristics in comparison to oil-gas results presented previously. As above, only FIN signal acquired by hydrophone #1 is presented, as the other three hydrophones recorded similar response.

Figure 8 demonstrates dependence of average power of acoustic FIN signal P_S on the total flowrate and water cut increasing from 0 to 100%. For any considered value of WC, a strong monotonic increase in the average FIN power is observed with increasing total flow rate (Figure 8, left). However, at any given flowrate, FIN power did not reveal any practical dependence on WC value (Figure 8, right).

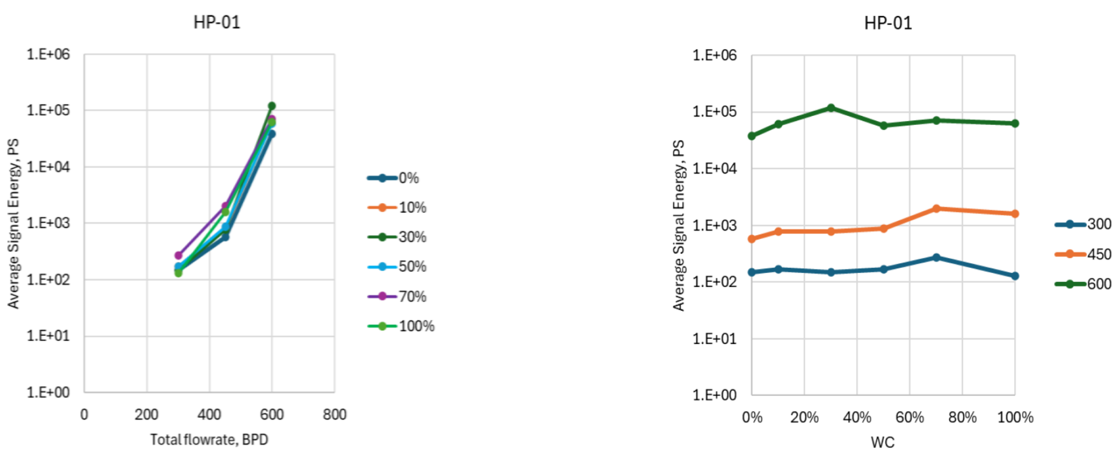

Figure 8—Oil-water test series: correlation of acoustic FIN signal energy on total flowrate and water cut (WC)

Furthermore, FIN spectral characteristics turned out to be practically insensitive to water breakthrough. In fact, oil-water spectrograms in Figure 9 show very similar with increasing WC. Unlike the previous case with oil and gas, the introduction of water into the flow does not result in the appearance of vertical "bands" in spectrograms to be related with intervals where water plugs interrupt the continuous flow of oil through the ICD. Time averaged spectrograms in Figure 10 reveals general match in frequency peaks between different water cuts. We attribute this low sensitivity of the FIN spectra properties to relatively low contrast in water and oil densities.

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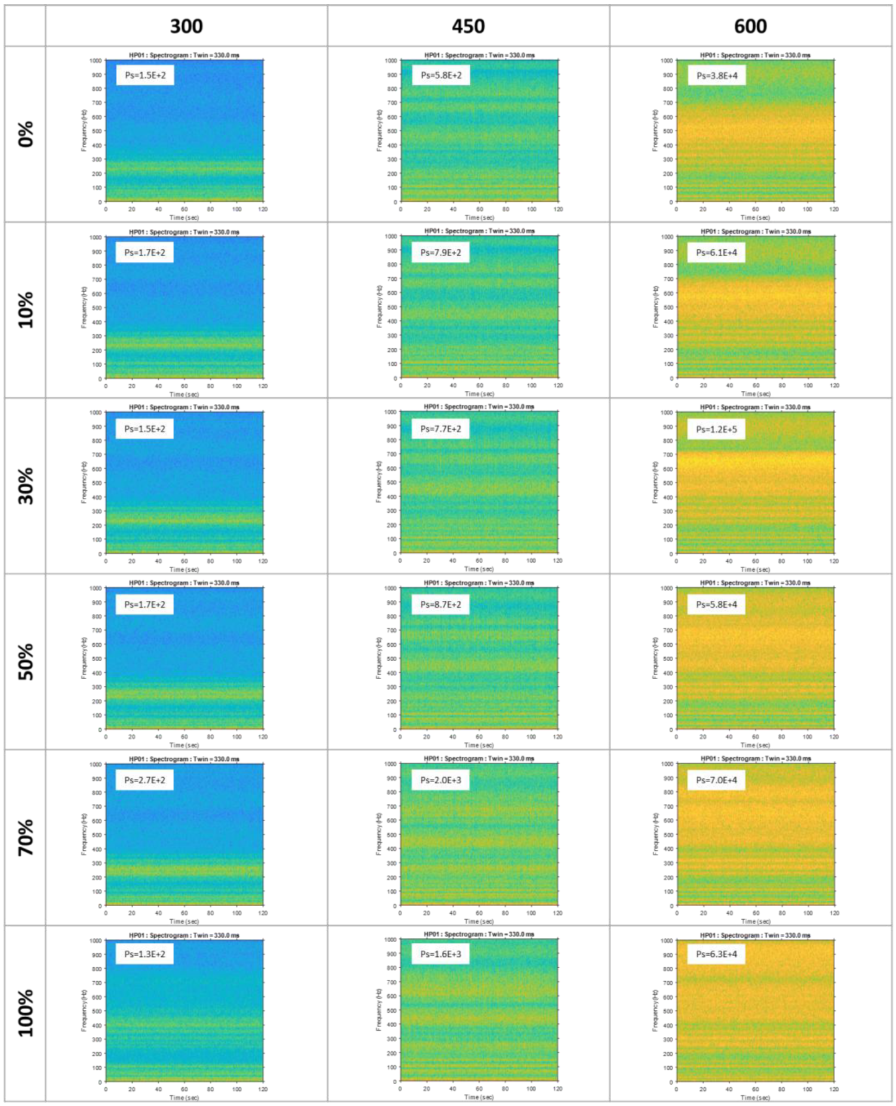

Figure 9—Oil-water test series: spectrograms of acoustic FIN acquired at various total flowrates and WC.

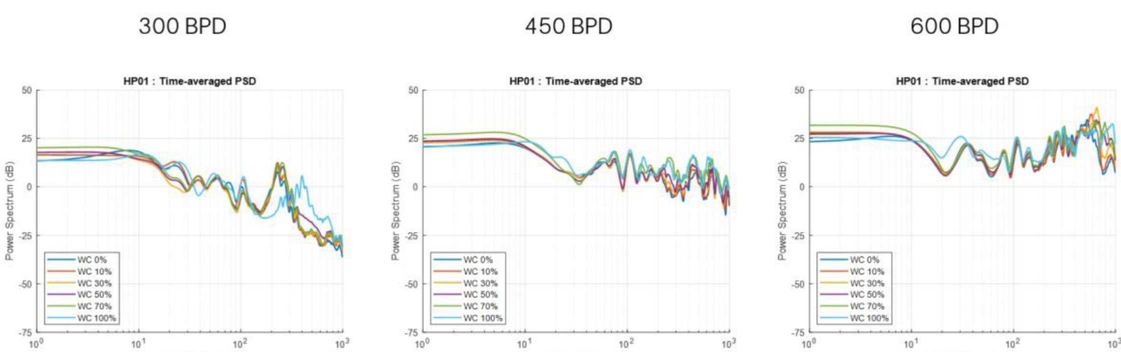

Figure 10—Oil-water test series: comparison of time-averaged spectrograms for various rates and GVF.

Conclusions

In this paper, feasibility of novel downhole acoustic flow monitoring technique in ICD configuration has been studied via large-scale 2-phase flow loop testing, involving ICD nozzle configuration. Experimental methodology was detailed and features the following:

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- Continuous fluid phase:
  - Dearomatized hydrocarbon solvent D80 as a lab proxy for the light crude oil
- Dispersed phase:
  - Water
  - Compressed air
- Water cut and gas volume fraction varied in discrete steps between 0 and 100%
- A wide range of flow rates was tested covering cavitation and non-cavitating conditions in the ICD.

Feasibility of determination multiphase flow characterization from FIN energy and spectral properties has been discussed. Overall, we observed different degree of sensitivity between multiphase flow characteristics (rate, phase volume fractions) and energy and spectral characteristics of acquired FIN.

In summary, we could see the following promising tendencies:

- GAS
  - Breakthrough even of small GVF is manifested by significant increase in acoustic signal power. This makes detection of gas breakthrough highly feasible.
  - Also, we saw significant variation of energy spectra of FIN with change of GVF, which makes promising detection of GVF. We recommend employing data analytics methods for to automate categorization of FIN based on flow characteristics.
- WATER
  - We did not observe significant changes in FIN power due to appearance of water in the flow. This makes detection of water breakthrough difficult.
  - Sensitivity of FIN spectral characteristics to changes in water cut was also found very week, regardless of flowrate magnitude. This makes detection of WC difficult.

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